

Table 1: CPU time on dataset collection 2 with 30.0% constraints (without centroid mutation)

| Constraints Level<br>Method<br>Dataset | 0.3  |        |      |
|----------------------------------------|------|--------|------|
|                                        | KSKM | KSKM_E | PCCC |
|                                        |      |        |      |
| synthetic_n300_d2_k10_s10              | 0.00 | 0.01   | 0.06 |
| synthetic_n300_d2_k10_s20              | 0.00 | 0.01   | 0.07 |
| synthetic_n300_d2_k10_s30              | 0.00 | 0.01   | 0.08 |
| synthetic_n300_d2_k10_s40              | 0.00 | 0.01   | 0.07 |
| synthetic_n300_d2_k10_s50              | 0.00 | 0.02   | 0.07 |
| synthetic_n300_d2_k20_s10              | 0.01 | 0.18   | 1.38 |
| synthetic_n300_d2_k20_s20              | 0.01 | 0.37   | 1.68 |
| synthetic_n300_d2_k20_s30              | 0.01 | 0.51   | 1.93 |
| synthetic_n300_d2_k20_s40              | 0.01 | 0.53   | 2.21 |
| synthetic_n300_d2_k20_s50              | 0.01 | 0.54   | 2.36 |
| synthetic_n300_d2_k50_s10              | 0.01 | 1.70   | 6.24 |
| synthetic_n300_d2_k50_s20              | 0.01 | 1.94   | 8.80 |
| synthetic_n300_d2_k50_s30              | 0.01 | 2.38   | 7.44 |
| synthetic_n300_d2_k50_s40              | 0.01 | 2.06   | 9.36 |
| synthetic_n300_d2_k50_s50              | 0.01 | 2.27   | 9.89 |